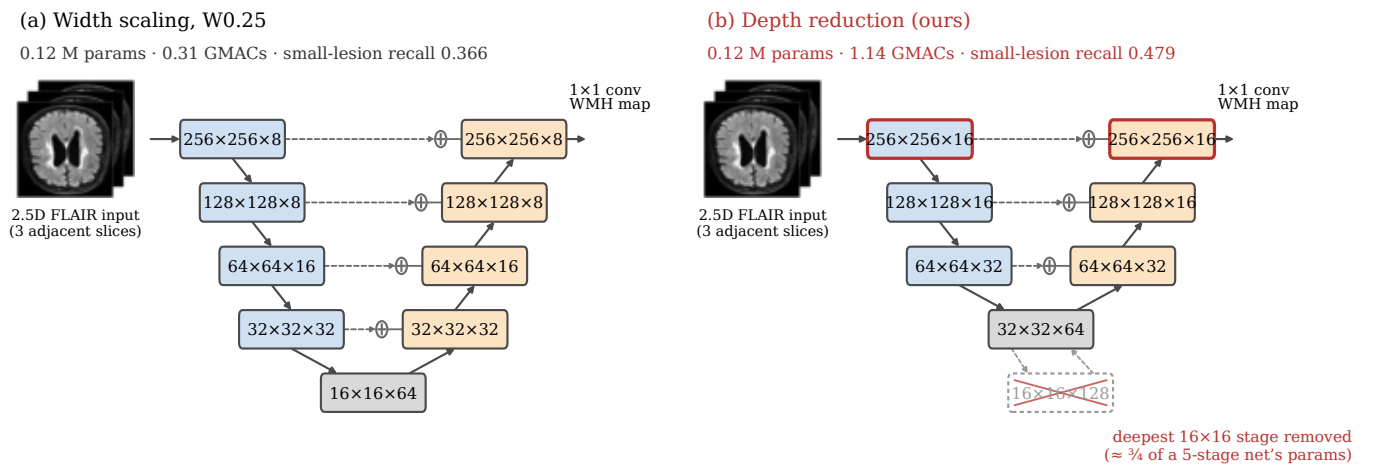
(a) Width scaling, W0.25
0.12 M params · 0.31 GMACs · small-lesion recall 0.366
2.5D FLAIR input
(3 adjacent slices)
256×256×8
128×128×8
64×64×16
32×32×32
16×16×64
256×256×8
128×128×8
64×64×16
32×32×32
1×1 conv
WMH map
all five stages kept;
every stage narrowed to 0.25× the baseline
encoder block: 2× (conv 3×3 – IN – LeakyReLU)
decoder block: up-conv, concat, 2× conv
(b) Depth reduction (ours)
0.12 M params · 1.14 GMACs · small-lesion recall 0.479
2.5D FLAIR input
(3 adjacent slices)
256×256×16
128×128×16
64×64×32
32×32×64
16×16×128
256×256×16
128×128×16
64×64×32
1×1 conv
WMH map
deepest 16×16 stage removed
(≈ ¾ of a 5-stage net's params)
saved budget moved to the high-resolution
stages (2× the channels of W0.25)
bottleneck (deepest stage)
skip connection (channel concatenation)